

Agentic AI is architecturally different

Traditional AI systems are reactive. You send a query; the model returns a response. The interaction is bounded, synchronous, and human initiated. Most enterprise AI today — chatbots, classification pipelines, recommendation engines — fits this pattern, and the architecture that supports it is essentially the architecture enterprises already have — APIs, databases, and request-response integration.

Agentic AI breaks this model in four meaningful ways. Agents initiate actions proactively, responding to environmental signals rather than waiting for human prompts. They orchestrate multi-step workflows autonomously across complex processes. They adapt their behaviour based on changing conditions rather than executing fixed decision trees. And they coordinate with each other — not just

with humans or APIs — to accomplish objectives that no single agent could complete alone.

The EY Technology Pulse Poll (April 2025), surveying more than 500 US technology leaders at organisations with 5,000 or more employees, found that 48% are already adopting or fully deploying agentic AI, with half expecting most of their AI to be autonomous within 24 months. The appetite is clearly there. The architectural readiness, in most cases, is not.

The numbers from Deloitte's 2025 Emerging Technology Trends study — based on a survey of 500 US technology leaders — are even more revealing about the gap between intent and reality: while 38% of organisations are piloting agentic solutions, only 11% are actively using these systems in production. A further 35% have no formal agentic strategy at all. Enthusiasm is not the

constraint. Infrastructure is.

The integration gap nobody discusses

Enterprise integration has been dominated by APIs for over a decade, and for good reason. APIs are well-understood, well-tooled, and excellent at what they do: exposing system functionality and data on demand. The problem is that 'on demand' is precisely the wrong model for autonomous agents.

Agentic systems need to know when something happens, not just retrieve data when asked. That gap — between point-in-time access and real-time awareness — is where most agentic deployments break down. Deloitte's Tech Trends 2026 report identifies legacy system integration as the primary infrastructure obstacle, noting that most agents still rely on

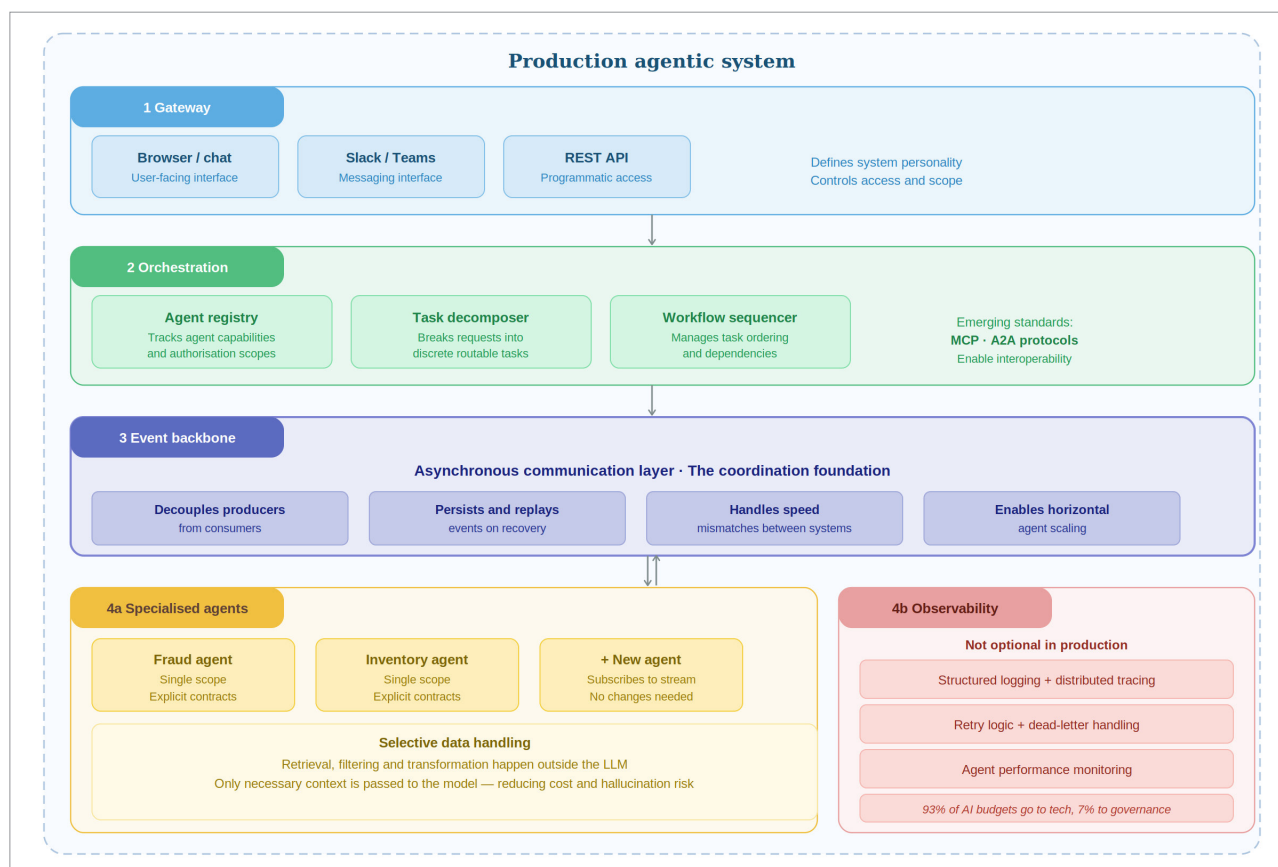


Figure 1: API-only vs event-driven architecture for agentic AI (Source: Deloitte Tech Trends 2026; Gartner (2025))